

Event-Driven Telemetry Platform – Project Timeline

This document outlines a realistic, resume-oriented timeline for building the Event-Driven Telemetry Platform. The focus is on backend architecture, correctness, and clarity rather than speed.

Phase	Duration	Goals	Key Deliverables
Phase 0	1–2 days	Set up local infrastructure and validate the Kafka pipeline.	Docker Compose, Kafka (KRaft), topic creation, CLI produce/consume validation.
Phase 1	3–4 days	Build a stateless HTTP ingestion service that publishes telemetry events.	FastAPI /track endpoint, schema validation, mocked auth, async Kafka producer.
Phase 2	2–3 days	Persist telemetry events reliably in a durable data store.	Kafka consumer, idempotent MongoDB writes, indexes on event_id and timestamp.
Phase 3	2–3 days	Maintain real-time metrics and counters for live observability.	Redis consumer, TTL-based counters, active users and event rate metrics.
Phase 4	1–2 days	Expose telemetry and metrics data via HTTP APIs.	GET /metrics endpoints reading from Redis, basic filtering.
Phase 5	1–2 days	Simulate traffic and demonstrate system behavior under load.	Event simulator script, controlled load testing, observation of metrics.
Phase 6	1–2 days	Polish the project and make it resume- and interview-ready.	README, architecture diagrams, setup instructions, demo narrative.

Total Estimated Time: ~12–16 focused days. This can be completed comfortably over 3–4 weeks alongside job preparation.

Each phase builds a complete vertical slice of the system, demonstrating backend engineering maturity rather than surface-level features.